

HEART FAILURE AND CARDIOMYOPATHIES

CLINICAL CASE

Isolated Atrial Myocarditis With Atrial Standstill



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ABSTRACT

BACKGROUND Myocarditis typically affects ventricles, whereas isolated atrial myocarditis is rare, causing conduction abnormalities, atrial standstill, and right heart failure.

CASE SUMMARY Two weeks after a respiratory infection, a 35-year-old man developed dyspnea, leg edema, and bradycardia. Electrocardiogram revealed a narrow QRS escape rhythm without P waves, and N-terminal pro-B-type natriuretic peptide was elevated. Echocardiography showed severe right ventricular dysfunction with massive tricuspid regurgitation. Cardiac magnetic resonance demonstrated atrial late gadolinium enhancement and mild edema, whereas electrophysiological testing confirmed atrial standstill and atrioventricular block.

DISCUSSION This case highlights isolated atrial myocarditis as a rare cause of atrial dysfunction, marked by postcapillary pulmonary hypertension and prominent V waves, despite normal mitral valves. Rapid atrial fibrosis progression resulted in silent atria, confirmed by magnetic resonance imaging and electrophysiology. Further research is required to clarify pathophysiology and establish effective management.

TAKE-HOME MESSAGES Isolated atrial myocarditis is a rare cause of atrial standstill and atrioventricular block. Comprehensive multimodality imaging is essential for accurate diagnosis and management. (JACC Case Rep. 2025;30:104029) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Myocarditis typically involves the ventricles; however, isolated atrial myocarditis is an uncommon clinical condition that has rarely been reported in the literature.^{1,2} It can present with complications, including conduction abnormalities, atrial standstill, and progression to right

heart failure. Due to its rarity and atypical clinical features, isolated atrial involvement poses significant diagnostic and management challenges. This case report highlights the clinical implications and distinctive hemodynamic features of isolated atrial myocarditis with atrial standstill.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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ABBREVIATIONS AND ACRONYMS

AV = atrioventricular
hs-TnT = high-sensitivity
troponin T

CASE PRESENTATION

A 35-year-old White man presented to the emergency department with dyspnea on mild exertion (NYHA functional class III), leg edema, and abdominal pain. He denied experiencing chest pain or dizziness. Two weeks prior, he reported symptoms consistent with a viral infection but decided to proceed with a planned trip. However, the progression of his symptoms led him to return early and seek medical attention. His medical history included ulcerative colitis, which had been well controlled with mesalazine, and a history of tobacco use.

INVESTIGATION

On physical examination, notable findings included peripheral edema in the lower extremities and bradycardia with a heart rate of 42 beats/min. The electrocardiogram demonstrated the absence of P waves with a narrow QRS escape rhythm (Figure 1). Laboratory findings showed an elevated N-terminal pro-B-type natriuretic peptide level of 432 pg/mL

TAKE-HOME MESSAGES

- Isolated atrial myocarditis is a rare clinical entity that can lead to complications, including conduction abnormalities, atrial standstill, and rapid progression to right heart failure.
- Comprehensive diagnostic evaluation, including cardiac magnetic resonance and electrophysiological testing, is crucial for accurate diagnosis and localization of fibrosis.
- Postcapillary pulmonary hypertension with prominent V waves in the absence of mitral valve abnormalities may be distinctive hemodynamic hallmarks of isolated atrial involvement.

(reference range: ≤ 125 pg/mL) and a high-sensitivity troponin T (hs-TnT) level of 31 ng/L (reference range: ≤ 14 ng/L). Mild elevations in liver enzymes and inflammatory markers were also noted. Trans-thoracic echocardiography showed a normally sized left ventricle with preserved systolic function and

FIGURE 1 Initial ECG



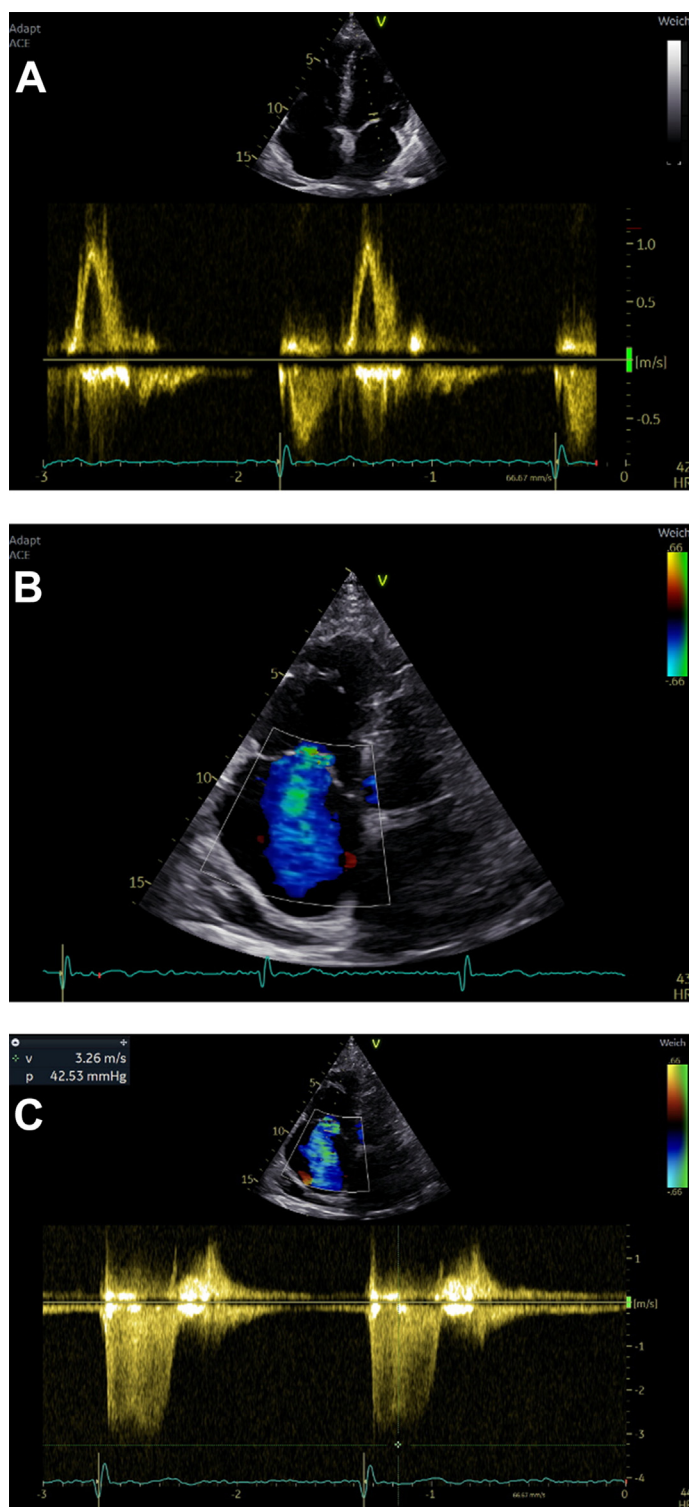
Initial ECG with an absence of P waves and narrow QRS complex with an escape rhythm at a bradycardic rate of 42 beats/min.
ECG = electrocardiogram.

mild mitral regurgitation. However, the left atrium and right ventricle were dilated, with evidence of pressure overload (D-sign), impaired right ventricular systolic function, and severe tricuspid regurgitation. The estimated systolic pulmonary artery pressure was elevated at 58 mm Hg (**Figure 2**). Given the clinical presentation and findings, a pulmonary embolism was initially suspected. The D-dimer level was markedly elevated at 2.94 $\mu\text{g/mL}$ (reference range: $<0.5 \mu\text{g/mL}$). However, computed tomography pulmonary angiography ruled out pulmonary embolism and instead revealed a dilated right ventricle, bilateral pleural effusions, and contrast agent reflux into the hepatic veins. To exclude the possibility of a right ventricular infarction, coronary computed tomography angiography was performed but showed only minimal stenosis of the proximal left anterior descending artery.

Right heart catheterization revealed a markedly elevated V-wave, reaching 65 mm Hg in the pulmonary arterial wedge pressure tracing without clearly definable A waves (**Figure 3**). This exceptionally high V-wave extended visibly into the pulmonary artery tracing, suggesting passive upstream pressure transmission of elevated left atrial pressure. Further hemodynamic assessment showed elevated mean pulmonary artery pressure at 35 mm Hg and mean pulmonary arterial wedge pressure at 28 mm Hg. The cardiac output was 5.8 L/min, resulting in a pulmonary vascular resistance that was normal at 1.2 WU, consistent with isolated postcapillary pulmonary hypertension associated with left heart disease.

Given the significant V-wave elevation, mitral disease was strongly suspected. However, initial transthoracic echocardiography revealed only mild mitral regurgitation, prompting further investigation. Transesophageal echocardiography was subsequently performed, confirming the transthoracic echocardiography findings and demonstrating only mild mitral regurgitation, effectively ruling out severe mitral valve pathology as the primary cause of the pronounced V-wave. A silent left atrium was considered as a rare cause of the giant V waves. Hence, cardiac magnetic resonance was performed, revealing normal left and right ventricular function but borderline elevated T_1 and T_2 times suggestive of mild myocardial edema. Late gadolinium enhancement was observed in the left atrium, consistent with fibrotic changes, whereas no significant scarring was observed in the ventricles (**Figure 4**). Based on the presence of 2 of 3 Lake Louise criteria (late gadolinium enhancement and myocardial edema) and elevated hs-TnT concentrations, a clinical diagnosis of myocarditis was made.³ Notably, in this case,

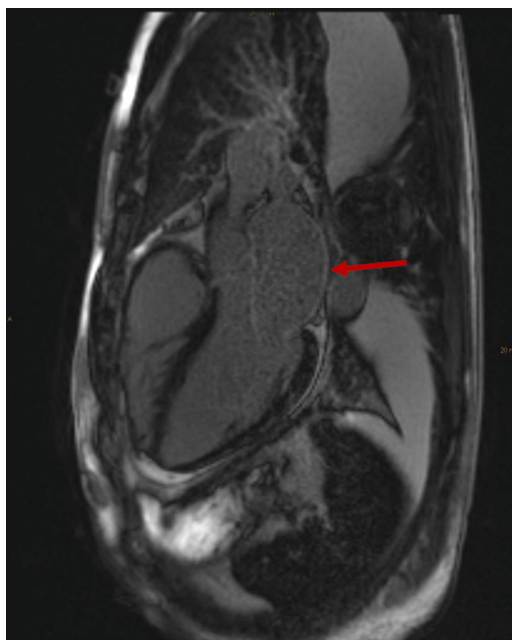
FIGURE 2 Transthoracic Echocardiography Performed at Admission



Transthoracic echocardiography performed at admission (A) with no visible A-wave in the mitral inflow Doppler image. (B) Right ventricle is dilated and shows severe systolic dysfunction with massive tricuspid regurgitation. (C) Estimated systolic pulmonary artery pressure is 58 mm Hg.

FIGURE 3 Right Heart Catheterization

Right heart catheterization displaying markedly elevated V waves in the PAWP. The tracing also demonstrates upstream pressure transmission to the PA. PA = pulmonary artery; PAWP = pulmonary arterial wedge pressure waveform.

FIGURE 4 Cardiac Magnetic Resonance

Cardiac magnetic resonance showing late gadolinium enhancement in the left atrium (red arrow, consistent with fibrotic changes).

myocarditis, at least in the postacute phase, appeared to be confined to the atria.

MANAGEMENT

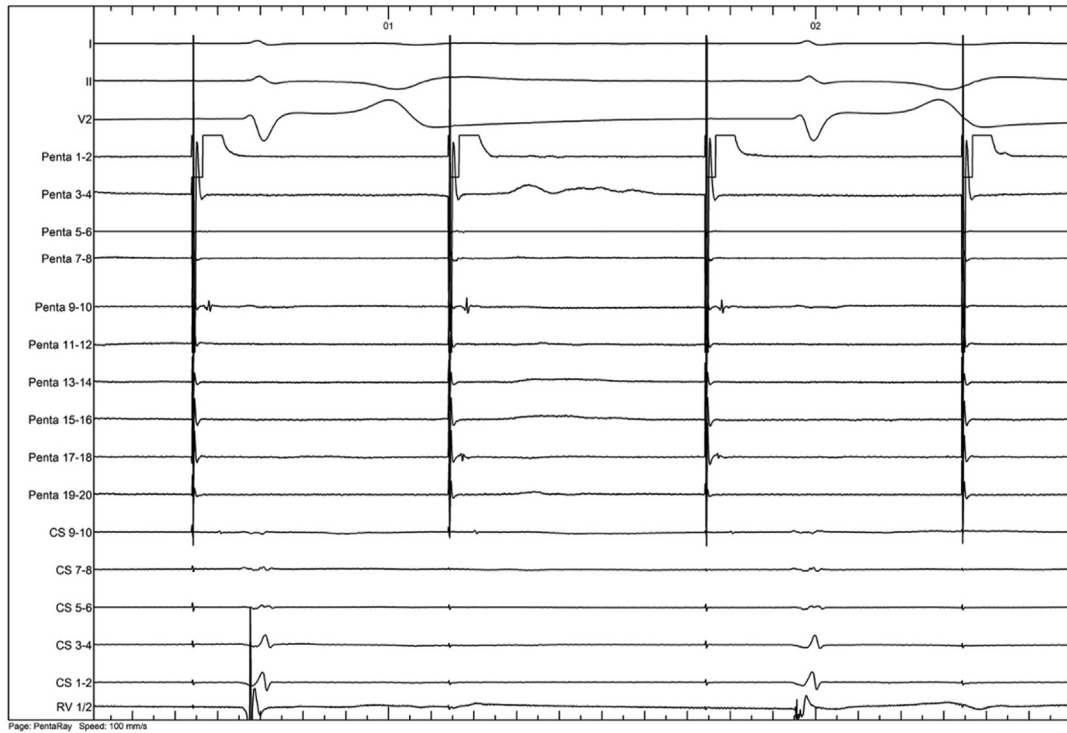
The patient was treated with furosemide and eplerenone, leading to symptomatic improvement and a decline in cardiac biomarker levels. Four days after admission, he developed dizziness and was managed with isoprenaline, which stabilized his heart rate. Due to a nonfunctioning atrioventricular (AV) node, a cardiac resynchronization therapy pacemaker was implanted because His bundle pacing was not available at that time. During the implantation, we attempted atrial pacing at multiple sites; however, all efforts were unsuccessful. Postinterventional transthoracic echocardiography showed the absence of an A-wave in the mitral inflow signal, a mildly reduced right ventricular systolic function, and massive tricuspid regurgitation.

OUTCOME AND FOLLOW-UP

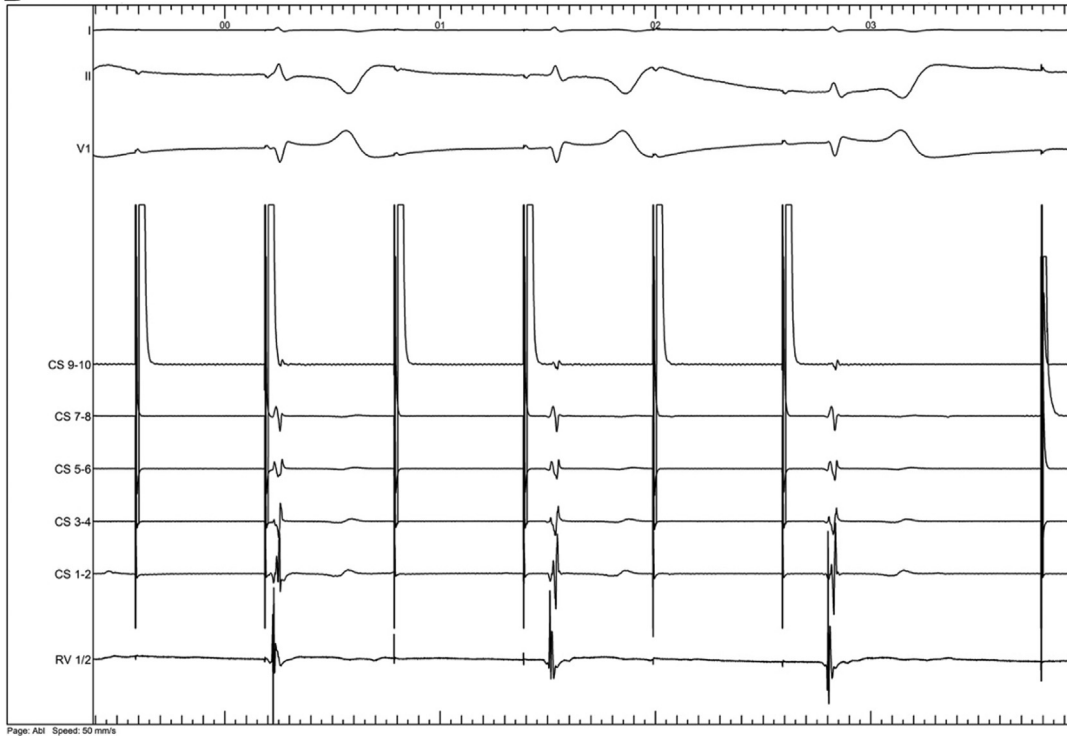
During his hospital stay, the patient lost a total 7 kg and was discharged in a recompensated state. Two weeks later, the patient was electively readmitted for an electrophysiological examination to evaluate the

FIGURE 5 Electrograms During Electrophysiological Examination

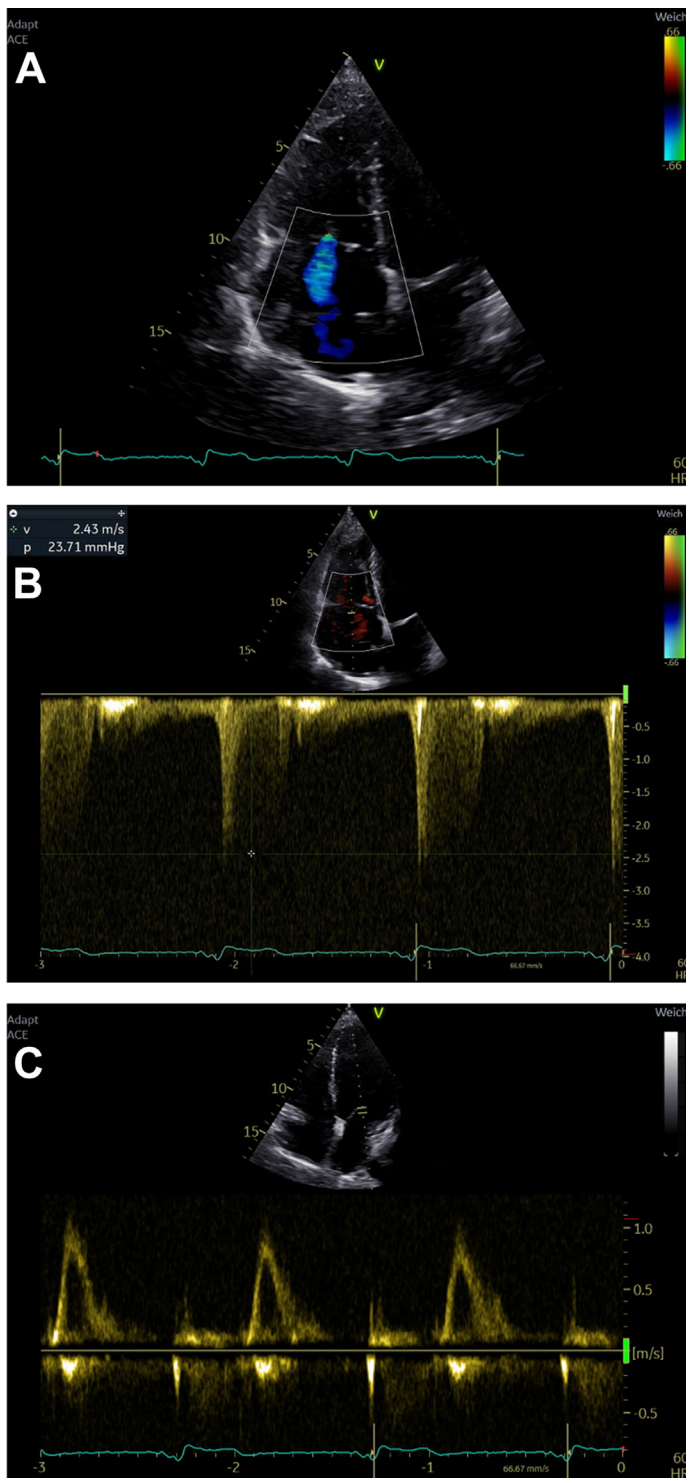
A



B



(A) Electrograms demonstrating septal pacing attempts via a pentaspline multipolar mapping catheter. Despite successful capture of the local septal atrial myocardium, as seen on Penta 9-10, pacing consistently resulted in complete atrioventricular block. (B) A Decapolar mapping catheter (CS) (Abbott) was inserted in the coronary sinus with the proximal bipole (CS 9-10) positioned at the coronary sinus ostium. The underlying rhythm is a junctional escape rhythm with narrow QRS complexes on the surface electrocardiogram. The CS shows only far-field ventricular signals without evidence of atrial activity. Additionally, pacing from the proximal CS bipole (CS 9-10) does not result in atrial capture.

FIGURE 6 Follow-Up Transthoracic Echocardiography Performed 6 Weeks Postdischarge

(A) Right ventricle is mildly dilated but shows normal systolic function. Tricuspid regurgitation is reduced, and (B) estimated systolic pulmonary artery pressure has improved to 29 mm Hg. (C) Left atrial activity remains absent.

potential benefit of relocating the atrial lead. Laboratory findings showed an hs-TnT level of 11 ng/L and an N-terminal pro-B-type natriuretic peptide level of 95 pg/mL. Electrophysiological mapping of the right atrium revealed a silent right atrium, with only a small atrial septal capture and complete AV block (Figure 5A). Furthermore, coronary sinus pacing showed no electrical response (Figure 5B). Consequently, repositioning the atrial lead was deemed unnecessary because septal stimulation resulted in no meaningful activity of the atria and complete AV block. Given a CHA₂DS₂-VA score of 1 (due to atherosclerosis) in the absence of atrial activity, anticoagulation was deferred, with plans for interval reassessment. At a clinical follow-up visit 6 weeks later, transthoracic echocardiography continued to show no left atrial activity. However, the right ventricle was only mildly dilated, with normal right ventricular systolic function, mild tricuspid regurgitation, and normal systolic pulmonary artery pressure (Figure 6). One month later, device interrogation confirmed 100% biventricular pacing.

DISCUSSION

This case report describes a rare presentation of isolated atrial myocarditis with atrial standstill leading to AV block and right ventricular dysfunction in a 35-year-old man. The rapid progression of the disease and unusual distribution of fibrosis confined to the atria make this case particularly noteworthy. Myocarditis typically involves the ventricles and can lead to ventricular arrhythmias or conduction abnormalities, and heart failure.^{2,4} In this case, the fibrosis was exclusively confined to the atria, with the ventricles completely unaffected. However, because only postmyocarditis changes were observed, it remains uncertain whether the acute phase of myocarditis was also restricted to the atria. Isolated atrial myocarditis has been rarely described in the literature and has been linked to giant cell myocarditis.¹ An endomyocardial biopsy was not performed in this case because cardiac magnetic resonance did not indicate active myocarditis, and a biopsy would not have influenced the therapeutic approach.

The rapid progression of myocarditis to fibrosis was another striking feature of this case. Despite the patient's recent viral infection and subsequent symptoms, fibrosis developed unusually quickly, leading to atrial dysfunction and complete, irreversible AV block. Such rapid progression of atrial myocarditis to fibrosis is atypical and considered rare.⁵ Although atrial fibrosis typically presents with atrial fibrillation or other supraventricular

arrhythmias,¹ this patient demonstrated a rare presentation of atrial inactivity in electrophysiological testing (silent atria).^{6,7} Another distinctive feature of this case was the presence of prominent V waves, despite the absence of mitral valve abnormalities. The rapid progression of atrial fibrosis ultimately contributed to the development of right heart failure. The absence of other contributing factors (eg, pulmonary embolism) suggests a pivotal role of atrial dysfunction in the pathogenesis of right heart failure, likely mediated by acute-onset postcapillary pulmonary hypertension caused by increased left atrial pressure.

This case highlights several important clinical implications: First, isolated atrial myocarditis should be considered as a rare differential diagnosis in patients presenting with unexplained AV block, particularly in the presence of recent viral infection or other systemic symptoms. Comprehensive diagnostic evaluation, including cardiac magnetic resonance and electrophysiological testing, is crucial for accurate diagnosis and localization of fibrosis. Postcapillary pulmonary hypertension with prominent V waves in the absence of mitral valve abnormalities may be a hallmark of isolated atrial involvement. Timely management with diuretics can result in significant clinical improvement. Although evidence is limited, consideration should also be given to anticoagulation

and antifibrotic therapies, including mineralocorticoid receptor antagonists, renin-angiotensin-aldosterone system inhibitors, and sodium-glucose cotransporter-2 inhibitors.

CONCLUSIONS

This case of isolated atrial myocarditis highlights an unusual distribution of postmyocarditis scarring confined to the atria, rapid progression to fibrosis, and resulting silent atria phenomenon with atrial standstill. Further research is necessary to understand the pathophysiology and establish effective management strategies for this rare condition.

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